

ExpressMart Sales Performance Analysis Project

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1. Introduction

This project aims to enhance overall business performance by understanding customer purchasing behaviour, identifying sales trends and optimizing operational efficiency. By understanding all the above, ExpressMart can gain several benefits.

Firstly, it can tailor its marketing strategies and promotions to target specific customer preferences, leading to increased sales and customer loyalty.

Secondly, identifying sales trends helps in inventory management, ensuring that popular products are always in stock while minimizing overstock of less popular items.

Lastly, optimizing operational efficiency can reduce costs, streamline processes, and improve overall profitability, enabling ExpressMart to stay competitive and responsive in a dynamic market environment.

The final dashboard contains visuals describing peak sales periods, top performing products based on sales and quantity ordered, regional sales performance and customer purchasing behavior. It also describes sales trends pertaining to the months, quarters and years as provided in the dataset.

2. Data Preparation

The ExpressMart sales data comprises sales information picked up from its point-of-sale systems. The dataset contains information on:

- Products sold, prices and the quantities ordered
- Order IDs and the order dates
- Purchase addresses and purchase times

There wasn't much to be done when it came to cleaning the data. There were no missing or duplicate values.

The data preparation process went as follows:

1. The data is loaded and formatted as a table.
2. The data is loaded into power query.
3. Order ID format is changed to text.
4. Order Date format is changed to date.
5. Time format is changed to time.
6. Price Each column is changed to Price.
7. Price and Sales format is changed to currency.
8. Hour column is created from the Time column.
9. Month column is created from the order date column.

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10. Quarter column is extracted from order date column and is formatted by adding the prefix “Qtr” to the cells.
 11. AM/PM column is extracted from the Time column and its format is changed to text.
 13. Year column is extracted from Order Date column and is formatted to text.
 14. City column is extracted from Purchase Address column.
 15. The cleaned dataset is loaded into a new worksheet.
 17. Pivot tables are created:
 - a. AM/PM vs Sum of Sales
 - b. Product vs Sum of Sales
 - c. City vs Sum of Sales
 - d. Quarter vs Sum of Sales
 - e. Year vs Sum of Sales
 - f. Hour vs Sum of Sales
 - g. Month vs Sum of Sales
 18. Total sales, total quantity ordered and total products were calculated.



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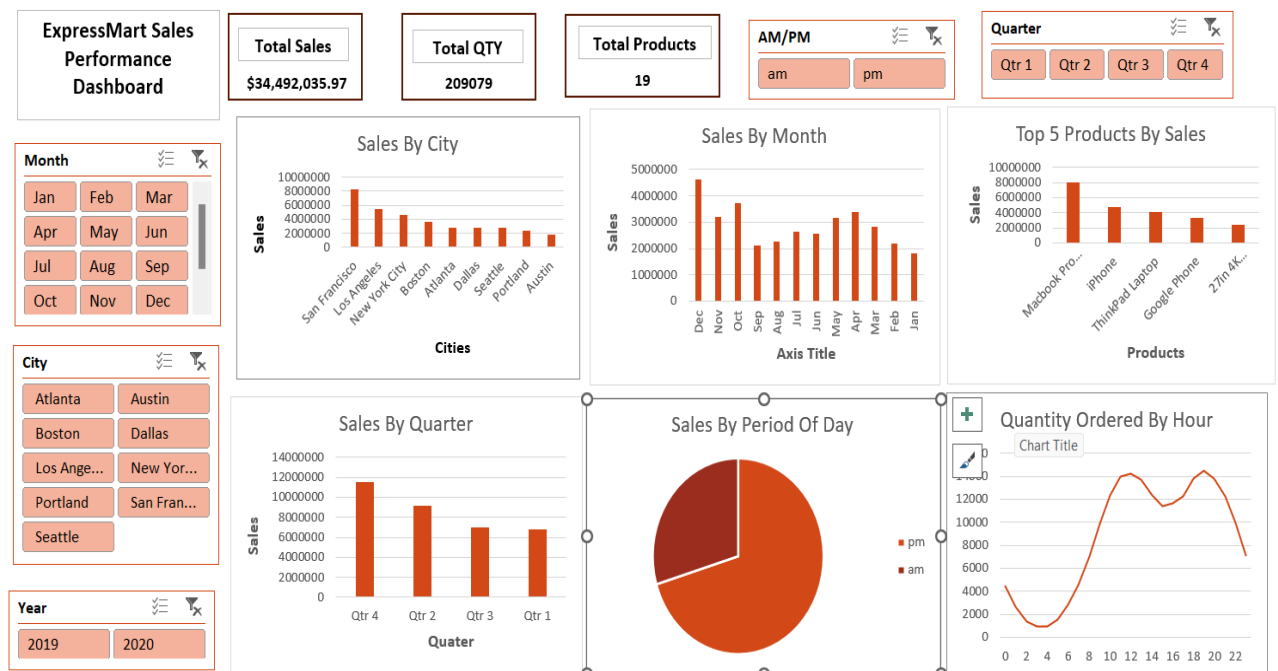
Row Labels	Sum of Sales	Row Labels	Sum of Sales	Row Labels	Sum of Sales	Row Labels	Sum of Sales
pm	24274253.98	Macbook Pro L	8037600	Qtr 4	11549773.42	2019	34483365.68
am	10217781.99	iPhone	4794300	Qtr 2	9121079.25	2020	8670.29
Grand Total	34492035.97	ThinkPad Laptc	4129958.7	Qtr 3	6989803.77	Grand Total	34492035.97
		Google Phone	3319200	Qtr 1	6831379.53		
		27in 4K Gaming	2435097.56	Grand Total	34492035.97		
Grand Total	22716156.26						
Row Labels	Sum of Quantity Ordered	Row Labels	Sum of Sales	Row Labels	Sum of Sales		
	4428	Dec	4613443.34	San Francisco	8262203.91		
1	2619	Oct	3736726.88	Los Angeles	5452570.8		
2	1398	Apr	3390670.24	New York City	4664317.43		
3	928	Nov	3199603.2	Boston	3661642.01		
4	937	May	3152606.75	Atlanta	2795498.58		
5	1493	Mar	2807100.38	Dallas	2767975.4		
6	2810	Jul	2647775.76	Seattle	2747755.48		
7	4556	Jun	2577802.26	Portland	2320490.61		
8	7002	Aug	2244467.88	Austin	1819581.75		
9	9816	Feb	2202022.42	Grand Total	34492035.97		
10	12308	Sep	2097560.13				
11	14005	Jan	1822256.73				
12	14202	Grand Total	34492035.97				
13	13685						
14	12362						
15	11391						
16	11662						
17	12239						
18	13802						
19	14470						
20	13768						
21	12244						
22	9899						
23	7065						
Grand Total	209079						

Order ID	Product	Quantity Ordered	Price	Order Date	Month	Year	Quarter	Purchase Address	City	Time	Hour	AM/PM	Sales	O	P
295665	Macbook Pro Laptop	1	1700	30/12/2019	Dec	2019	Qtr 4	136 Church St, New York City, NY 10001	New York City	12:01:00 AM	0	am	1700		
295666	LG Washing Machine	1	600	29/12/2019	Dec	2019	Qtr 4	562 2nd St, New York City, NY 10001	New York City	7:03:00 AM	7	am	600		
295667	USB-C Charging Cable	1	11.95	12/12/2019	Dec	2019	Qtr 4	277 Main St, New York City, NY 10001	New York City	6:21:00 PM	18	pm	11.95		
295668	27in FHD Monitor	1	149.99	22/12/2019	Dec	2019	Qtr 4	410 6th St, San Francisco, CA 94016	San Francisco	3:13:00 PM	15	pm	149.99		
295669	USB-C Charging Cable	1	11.95	18/12/2019	Dec	2019	Qtr 4	43 Hill St, Atlanta, GA 30301	Atlanta	12:38:00 PM	12	pm	11.95		
295670	AA Batteries (4-pack)	1	3.84	31/12/2019	Dec	2019	Qtr 4	200 Jefferson St, New York City, NY 10001	New York City	10:58:00 PM	22	pm	3.84		
295671	USB-C Charging Cable	1	11.95	16/12/2019	Dec	2019	Qtr 4	928 12th St, Portland, OR 97035	Portland	3:10:00 PM	15	pm	11.95		
295672	USB-C Charging Cable	2	11.95	13/12/2019	Dec	2019	Qtr 4	813 Hickory St, Dallas, TX 75001	Dallas	9:29:00 AM	9	am	23.9		
295673	Bose SoundSport Headphones	1	99.99	15/12/2019	Dec	2019	Qtr 4	718 Wilson St, Dallas, TX 75001	Dallas	11:26:00 PM	23	pm	99.99		
295674	AAA Batteries (4-pack)	4	2.99	28/12/2019	Dec	2019	Qtr 4	77 7th St, Dallas, TX 75001	Dallas	11:51:00 AM	11	am	11.96		
295675	USB-C Charging Cable	2	11.95	13/12/2019	Dec	2019	Qtr 4	594 1st St, San Francisco, CA 94016	San Francisco	1:52:00 PM	13	pm	23.9		
295676	ThinkPad Laptop	1	999.99	28/12/2019	Dec	2019	Qtr 4	410 Lincoln St, Los Angeles, CA 90001	Los Angeles	5:19:00 PM	17	pm	999.99		
295677	AA Batteries (4-pack)	2	3.84	20/12/2019	Dec	2019	Qtr 4	866 Pine St, Boston, MA 02215	Boston	7:19:00 PM	19	pm	7.68		
295678	AAA Batteries (4-pack)	2	2.99	06/12/2019	Dec	2019	Qtr 4	187 Lincoln St, Dallas, TX 75001	Dallas	9:38:00 AM	9	am	5.98		
295679	USB-C Charging Cable	1	11.95	25/12/2019	Dec	2019	Qtr 4	902 2nd St, Dallas, TX 75001	Dallas	9:39:00 AM	9	am	11.95		
295680	Lightning Charging Cable	1	14.95	01/12/2019	Dec	2019	Qtr 4	338 Main St, Austin, TX 73301	Austin	2:30:00 PM	14	pm	14.95		
295681	Google Phone	1	600	25/12/2019	Dec	2019	Qtr 4	79 Elm St, Boston, MA 02215	Boston	12:37:00 PM	12	pm	600		
295681	USB-C Charging Cable	1	11.95	25/12/2019	Dec	2019	Qtr 4	79 Elm St, Boston, MA 02215	Boston	12:37:00 PM	12	pm	11.95		
295681	Bose SoundSport Headphones	1	99.99	25/12/2019	Dec	2019	Qtr 4	79 Elm St, Boston, MA 02215	Boston	12:37:00 PM	12	pm	99.99		
295681	Wired Headphones	1	11.99	25/12/2019	Dec	2019	Qtr 4	79 Elm St, Boston, MA 02215	Boston	12:37:00 PM	12	pm	11.99		
295682	USB-C Charging Cable	1	11.95	23/12/2019	Dec	2019	Qtr 4	780 Elm St, Portland, OR 97035	Portland	7:25:00 PM	19	pm	11.95		
295683	Wired Headphones	1	11.99	23/12/2019	Dec	2019	Qtr 4	341 Lake St, San Francisco, CA 94016	San Francisco	10:46:00 PM	22	pm	11.99		
295684	AAA Batteries (4-pack)	4	2.99	04/12/2019	Dec	2019	Qtr 4	936 Church St, San Francisco, CA 94016	San Francisco	12:29:00 PM	12	pm	11.96		
295685	Wired Headphones	1	11.99	11/12/2019	Dec	2019	Qtr 4	662 Ridge St, San Francisco, CA 94016	San Francisco	10:54:00 PM	22	pm	11.99		
295686	USB-C Charging Cable	1	11.95	17/12/2019	Dec	2019	Qtr 4	573 Maple St, Portland, ME 04101	Portland	4:10:00 PM	16	pm	11.95		
295687	Apple AirPods Headphones	1	150	10/12/2019	Dec	2019	Qtr 4	531 Ridge St, Boston, MA 02215	Boston	12:38:00 AM	0	am	150		
295688	Apple AirPods Headphones	1	150	08/12/2019	Dec	2019	Qtr 4	546 Park St, Los Angeles, CA 90001	Los Angeles	11:21:00 AM	11	am	150		
295689	LG Washing Machine	1	600	24/12/2019	Dec	2019	Qtr 4	173 Lake St, San Francisco, CA 94016	San Francisco	11:21:00 AM	11	am	600		
295690	Apple AirPods Headphones	1	150	04/12/2019	Dec	2019	Qtr 4	174 Cedar St, Seattle, WA 98101	Seattle	12:44:00 PM	0	am	150		
295691	Bose SoundSport Headphones	1	99.99	22/12/2019	Dec	2019	Qtr 4	102 Johnson St, Los Angeles, CA 90001	Los Angeles	6:34:00 PM	18	pm	99.99		
295692	AA Batteries (4-pack)	2	3.84	14/12/2019	Dec	2019	Qtr 4	640 Cherry St, Los Angeles, CA 90001	Los Angeles	1:21:00 AM	1	am	7.68		

3. Dashboard Design and Features

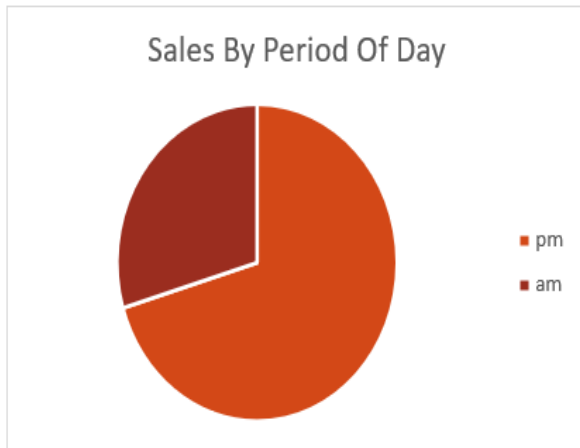
The ExpressMart sales performance dashboard seeks to visually expose trends in sales, peak sales periods, top-performing products, regional sales performance and customer purchasing habits. As such, it includes features that make it achieve its purpose effectively. These features are:

- Cards displaying the total sales, total quantity ordered and total products.
- Slicers for AM/PM, month, quarter, city and year.
- Clustered column charts for sales by city, sales by month, sales by quarter and top 5 products by sales.
- Pie chart for sales by period of day.
- Line chart for quantity ordered by hour.

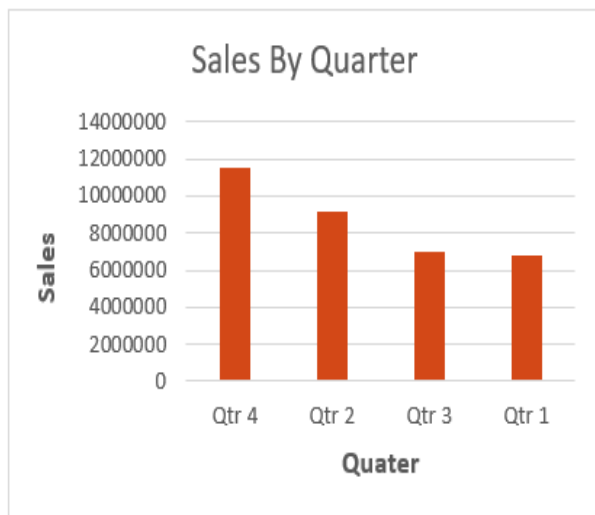


4. Key Insights

Peak Sales Period:

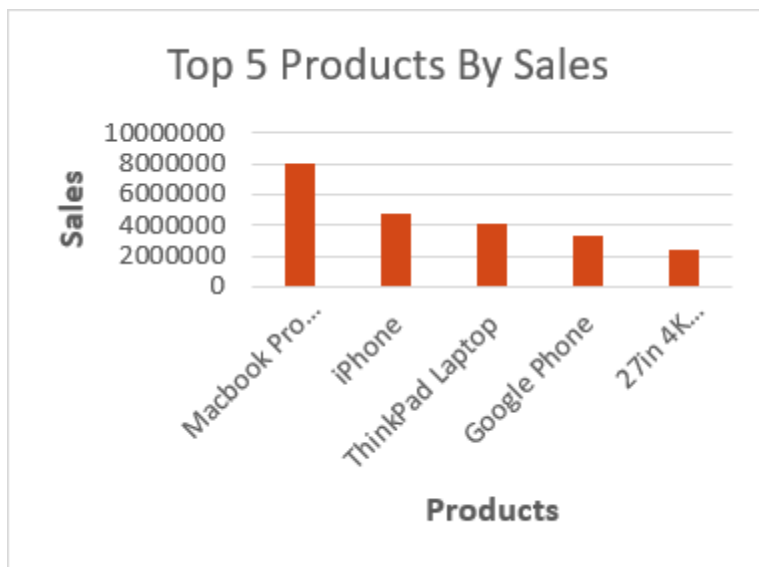


1. Sales peak during the PM, specifically in the afternoon at 12 and then again at 7pm when it peaks highest. 70% of sales occur during the PM period.



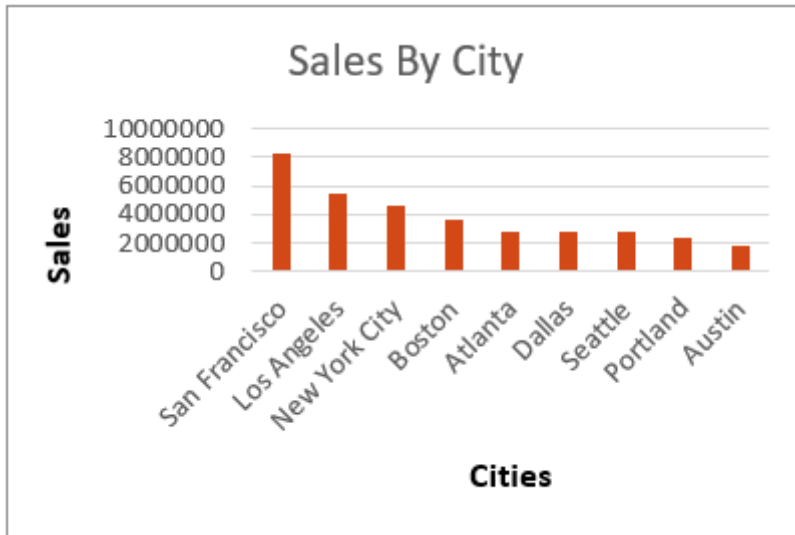
2. Sales peak during the 4th quarter in the months of October, November and December. This is because this quarter is marked by festive seasons which drive people to shop a lot during this time, thereby increasing sales to a significant degree.

Top-Performing Products:



1. The top-performing products in sales are:
 - a. MacBook Pro Laptop
 - b. iPhone
 - c. ThinkPad Laptop
 - d. Google Phone
 - e. 27in 4K Gaming Monitor
2. The top-performing products are electronic gadgets. This might be due to the fact that electronic gadgets are in high and constant demand these days.
3. Another reason for such performance is that these gadgets are expensive such that even selling a few units of them can make them top performers in sales.

Regional Sales Performance:

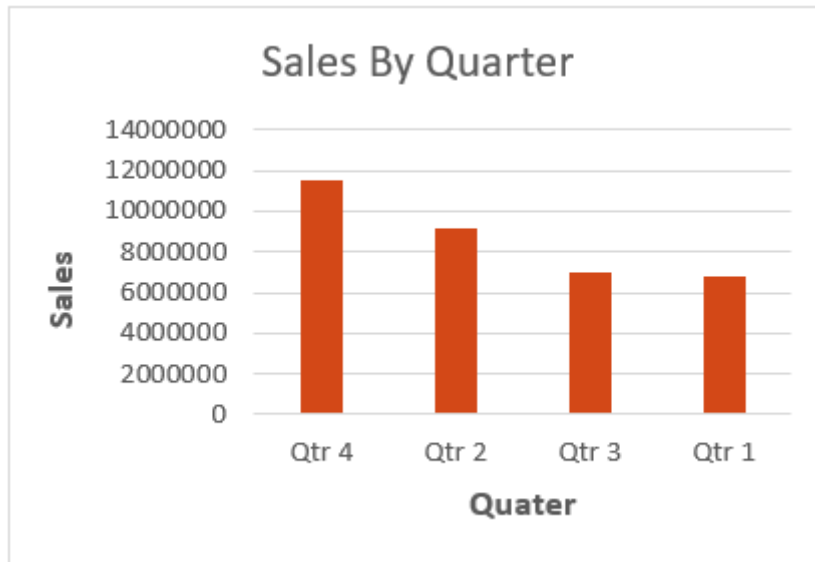


1. San Francisco is the highest-performing city, followed by Los Angeles and New York. The common factor between these 3 cities is that they are large cities which constitutes a large market. Thereby making them the top sales performers.
2. Portland and Austin are the worst-performing cities. In contrast to the top 3 cities, these are much smaller cities making them a small market as compared to the rest. This is the reason for their seemingly poor performance.

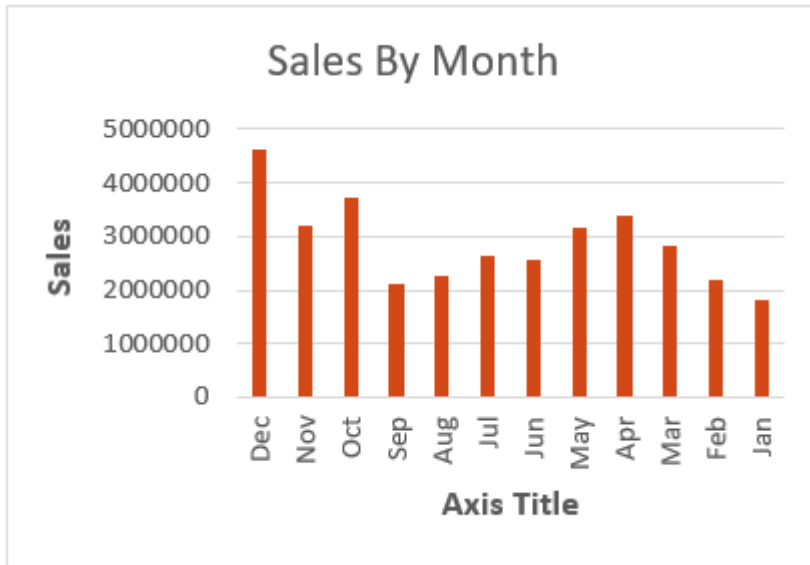
Customer Purchasing Patterns:

1. Customers' purchase rate peaks at noon and then a second time at 7 pm. This is because at 12 said customers might be on a lunch break, affording them the time to make purchases and by 7 pm most people would have already closed from work, also making it an ideal time to make purchases.
2. Customers' purchase rate is significantly higher in the 2nd and 4th quarters of the year. They prefer to shop more during such periods as there are festive seasons in these periods as well as holidays.

Sales Trend Analysis:



1. Sales are at their lowest in the 1st quarter, increasing in the 2nd quarter and then dipping in the 3rd quarter before peaking in the 4th quarter.
 - a. Sales are lowest in the 1st quarter because it is the beginning of the year and people are recuperating their expenses from the previous quarter in which they spent heavily.
 - b. The sudden increase in sales in the 2nd quarter can be attributed to the presence of festive seasons such as Easter.
 - c. The 3rd quarter sees a sales slump because that period is marked by fewer holidays and festive seasons.
 - d. The 4th quarter sees a significant boom in sales largely due to the fact it is marked by festive seasons such as Christmas and Thanksgiving.



2. Sales are at their lowest in January, increasing gradually till they peak for the first time in April, then regress till they peak again in October. Sales go down a bit in November before peaking highest in December.

5. Recommendations

1. During peak periods more staff must be stationed at the workplace to handle the huge influx of customers. Inventory is also to be increased among the various products by their purchase rate during these periods. This is to prevent understaffing during the peak periods whilst also ensuring efficient inventory management.
2. The poor-performing products are mostly those of significantly lower price value as compared to the top performers. As such, the poor-performing products could be attached to the top performers and be sold

as a package. This will improve the sales numbers of the poor-performing products.

3. Cities like Austin, Portland, Seattle, Dallas, Atlanta and Boston are not performing up to par but show great growth potential. Targeted ad campaigns must be issued in these cities, making the presence of ExpressMart more pronounced. This will drive in more customers thereby increasing sales.
4. Customers tend to shop more during festive seasons and holidays. As such, holiday sales promos and similar events can be held during said periods to drive more customer interest and boost sales even further.
5. During down periods, such as January, February, August, and September, marketing strategies such as discounted price campaigns and product package deals can be leveraged to increase sales.

6. Conclusion

ExpressMart Sales Performance Analysis Summary

- **Objective:** Improve business by understanding customer behaviour, sales trends, and operational efficiency.
- **Data:** Sales data, including product details, order dates, and purchase information, prepared and analysed.
- **Dashboard:** Visuals on sales trends, peak periods, top products, and regional performance.

Key Insights

- Sales peak in the PM and 4th quarter.
- Top products: high-demand electronics.
- Highest sales in large cities like San Francisco.
- Purchase rates are higher during festive seasons.

Recommendations

- Increase staff and inventory during peak times.
- Pair low-performing products with top-sellers.
- Targeted ads in smaller, growing cities.
- Run holiday promos and discounts during slow periods.

Future Work

Here are some suggestions for further analysis and potential enhancements to the dashboard:

Further Analysis:

1. **Sales Forecasting:** Implement predictive analytics to forecast future sales trends based on historical data. This can aid in better inventory planning and demand management.
2. **Competitive Analysis:** Compare sales performance with competitors to identify strengths, weaknesses, and opportunities for improvement.

Dashboard Enhancements:

1. **Real-time Data Updates:** Integrate real-time data feeds to keep the dashboard updated with the latest sales information, enabling quicker decision-making.
2. **Customer Feedback Integration:** Incorporate customer feedback data to correlate sales performance with customer satisfaction and identify areas for service improvement.

3. Performance Benchmarking: Include benchmarking features to compare current performance against set targets or industry standards.

Implementing these analyses and enhancements can provide deeper insights into sales performance, improve strategic decision-making, and drive overall business growth for ExpressMart.



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